

Digital Engineering

Course Overview

Course Description

- This course introduces students to the basic concepts of digital systems, including analysis and design. Both combinational and sequential logic will be covered. Students will gain experience with several levels of digital systems, from simple logic circuits to programmable logic devices.

Course Objectives

- Having successfully completed this course, the student will be able to:
  - Express the operation of logic gates with the aid of Boolean algebra
  - Minimize Boolean expression using laws and rules of Boolean algebra and the Karnaugh map
  - Analyze and design combinational and sequential logic circuits
  - Discuss the design concepts of basic memory elements and explain the operation of various memory types.
  - Programmable logic

Textbooks

- Main Textbook:
  - Morris Mano and Michael Ciletti, "Digital Design", Prentice Hall, 5th Edition, 2013.

- Additional References:
  - Thomas Floyd, "Digital Fundamentals", Prentice Hall, 10th Edition, 2008.
  - Ronald J. Tocci, “Digital Systems: Principles and Applications”, Prentice Hall, 11th Edition, 2010.

Tentative Course Outline

Week	Topic	Week	Topic
1	• Course Overview	8	• Synchronous and Asynchronous Sequential Circuits: Analysis and Design
2	• Introduction • Boolean algebra and logic gates	9	• Synchronous and Asynchronous Sequential Circuits: Analysis and Design (Cont'd)
3	• Boolean algebra and logic gates (Cont'd) • Simplification of Boolean functions	10	• Registers and Counters
4	• Simplification of Boolean functions (Cont'd)	11	• Registers and Counters (cont'd)
5	• Combinational logic circuits	12	• Memory devices
6	• Combinational logic circuits (Cont'd)	13	• Programmable Logic Device
7	• Midterm Exam I	14	• Review and Lab Exam